

- 1- A) To encrypt the data being sent.
B) To increase the power of transmitted signals.
C) To connect two devices for data transfer.
D) To establish security measures on a network.

2- At the network edge, which of the following typically acts as the primary initiators of communication?

- A) Routers
B) Clients
C) Servers
D) Switches

3- What is the advantage of using a full-duplex communication link?

- A) It allows data to be transmitted and received simultaneously.
B) It reduces the signal strength needed for transmission.
C) It allows data to be transmitted in only one direction at a time.
D) It decreases the overall bandwidth required for communication.

4- Which of the following is NOT a characteristic of a server in a network?

- A) It primarily initiates communication with clients.
B) It typically provides resources or services to clients.
C) It is usually more powerful and capable of handling multiple requests simultaneously.
D) It often resides in a data center for better scalability and availability.

5- Which of the following best describes the client-server model in networking?

- A) A system where every device can act as both a client and a server simultaneously.
B) A model where clients request services and servers respond with the requested resources.
C) A model based on peer-to-peer communication where no centralized server is required.
D) A model where clients provide services to other clients directly

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6- Which of the following physical media is commonly used in fiber-optic communication?

- A) Copper wires
B) Radio waves
C) Glass or plastic fibers
D) Microwaves

7- Which of the following is a characteristic of circuit switching?

- A) Data is broken into packets and sent independently.
- B) A dedicated path is established for the entire communication session.
- C) Resources are allocated dynamically based on traffic demand.
- D) It is more efficient than packet switching for burst traffic.

8- Which of the following is true about circuit switching?

- A) It is more efficient than packet switching for long-distance communication.
- B) It is more cost-effective than packet switching for high-volume traffic.
- C) It is better suited for applications with unpredictable data traffic patterns.
- D) It guarantees a fixed path for communication, reducing delays

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9- What is the main purpose of HTTP (Hypertext Transfer Protocol)?

- A) To encrypt data between web browsers and servers
- B) To transfer files over the Internet
- C) To enable communication between web browsers and web servers
- D) To manage email transmission

10. Which HTTP status code indicates a successful HTTP request?

- A) 200
- B) 301
- C) 500
- D) 403

11. What is the primary function of the HTTP header in a request or response?

- A) To provide content for the body of the request/response
- B) To specify the status of the HTTP request
- C) To provide additional metadata such as content type, encoding, and caching information
- D) To encrypt the transmitted data

12. What is the main difference between POP3 and IMAP?

- A) POP3 stores emails on the client side, while IMAP stores emails on the server.
- B) IMAP retrieves emails from the server and removes them, while POP3 leaves emails on the server.
- C) IMAP is used for sending emails, while POP3 is used for receiving emails.
- D) IMAP requires an internet connection, while POP3 does not.

13. SMTP is used primarily for which of the following?

- A) Retrieving email messages

- B) Sending email messages
- C) Storing email messages
- D) Managing email inboxes

14. Which email protocol does not allow email management on the server (such as organizing folders or marking emails as read), but instead downloads and removes the email from the server?

- A) SMTP
- B) IMAP
- C) POP3
- D) HTTP

15. What is a key advantage of IMAP over POP3?

- A) IMAP is used only for sending emails, while POP3 is used for receiving emails.
- B) IMAP allows emails to be accessed and organized from multiple devices while keeping them on the server.
- C) IMAP does not require an Internet connection, while POP3 does.
- D) IMAP automatically deletes emails from the server after download.

16. SMTP can use which of the following methods to secure email communication?

- A) Encryption with SSL/TLS
- B) Email compression
- C) Authentication with a username and password
- D) All of the above

17. Which of the following is a characteristic feature of Peer-to-Peer (P2P) applications?

- A) They rely on a central server to manage connections between peers.
- B) They allow direct communication between clients without the need for an intermediary.
- C) They are primarily used for cloud storage.
- D) They can only be used for file sharing.

18. Which of the following is a characteristic of UDP?

- A) It provides reliable data delivery and ensures that data is received in the correct order.
- B) It guarantees that no packets are lost during transmission.
- C) It is suitable for real-time applications like video streaming and online gaming.
- D) It requires a three-way handshake for establishing a connection.

19. Which of the following is the primary purpose of the Domain Name System (DNS)?

- A) To provide encryption for web traffic.
- B) To translate domain names into IP addresses.
- C) To ensure secure communication between email servers.
- D) To store the content of web pages.

20. Which of the following is the primary responsibility of the transport layer in the OSI model?

- A) Error detection and correction at the bit level
- B) Reliable end-to-end communication between hosts
- C) Routing packets across different networks
- D) Addressing devices on the same network

21. Which of the following protocols operates at the transport layer and ensures reliable data transmission?

- A) IP
- B) UDP
- C) TCP
- D) ARP

22. Which of the following is a key difference between TCP and UDP?

- A) TCP uses connectionless communication, while UDP uses connection-oriented communication.
- B) TCP guarantees data delivery and sequencing, while UDP does not.
- C) UDP is more secure than TCP.
- D) TCP is faster than UDP for all types of applications.

23- What does the term "port number" refer to in transport layer communication?

- A) A logical endpoint for communication used to differentiate between different services on a device.
- B) A physical location where data is received.
- C) A unique identifier for each network.
- D) The time duration for which a data connection is open.

24. Which of the following is a characteristic of a connection-oriented service?

- A) Data is delivered without any acknowledgment.
- B) Data is transferred without any order.
- C) The connection is established before data transfer begins.
- D) It is more suitable for time-sensitive applications like video streaming.

25. What is "congestion control" in the context of the transport layer?

- A) A mechanism to prevent buffer overflow at the receiving host.
- B) A mechanism to prevent excessive data loss by avoiding network congestion.
- C) A method to ensure data is delivered in sequence.
- D) A method to reduce errors during data transmission.

26. Which of the following best describes how demultiplexing works in UDP?

- A) Data is delivered to a specific process based on the port number specified in the UDP header.
- B) Data is forwarded to the appropriate IP address.
- C) The data is sent directly to the receiving host without using any identifiers.
- D) The server assigns a port number for each incoming connection.

27. What happens if a transport layer demultiplexing process receives a packet with an unknown destination port number?

- A) The packet is forwarded to the default application process.
- B) The packet is discarded by the transport layer.
- C) The packet is passed to the application layer for inspection.
- D) The packet is retransmitted to the source.

28. In a client-server model, if multiple clients are sending messages to a server, what helps the server determine which client the message came from?

- A) Source IP address and source port number
- B) Destination IP address
- C) Sequence number in the transport layer header
- D) Application layer protocol used

29. Which of the following is an essential feature of a reliable data transfer protocol?

- A) No error detection
- B) Fast transmission speed with no packet loss
- C) A method to ensure that data is delivered without loss, corruption, or duplication
- D) The ability to ignore sequence numbers and retransmit data when needed

30. What happens if a reliable data transfer protocol detects that a packet was lost?

- A) The protocol retransmits the lost packet.
- B) The protocol discards the packet and continues.
- C) The protocol waits until all packets are received.
- D) The protocol sends an acknowledgment for the missing packet.

31. What is the purpose of the three-way handshake in TCP connection management?

- A) To establish a reliable connection between the sender and receiver.
- B) To encrypt the data before sending it.
- C) To determine the maximum packet size for transmission.
- D) To route packets across multiple networks.

32. What is the purpose of sequence numbers in TCP connection management?

- A) To ensure that the connection is secure.
- B) To identify and reorder packets to ensure data integrity and correct order.
- C) To assign priority levels to packets.
- D) To encrypt data during transmission.

33. In the TCP connection termination process, how does the server indicate it is ready to close the connection?

- A) By sending a FIN packet.
- B) By sending a SYN-ACK packet.
- C) By sending a RESET packet.
- D) By sending an ACK packet.

34. What happens when TCP detects congestion in the network?

- A) It increases the size of the congestion window.
- B) It reduces the congestion window size.
- C) It sends more packets to test the network's capacity.
- D) It ignores the congestion and continues to send data.

35. What is the congestion window (cwnd) in TCP?

- A) The maximum number of packets that can be transmitted over the network.
- B) The maximum number of packets that the receiver can process at a given time.
- C) The maximum amount of data that can be sent before waiting for an acknowledgment.
- D) The size of the buffer used for storing packets before transmission.

36. In TCP's congestion control, what does the Threshold (ssthresh) represent?

- A) The point where the sender will switch from exponential to linear growth of the congestion window.
- B) The maximum value of the congestion window during the slow start phase.
- C) The point where the sender stops sending new packets.
- D) The maximum number of packets that can be in transit at any given time

Question_2:

2.1 Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end 1 delay?

2.2 What information is used by a process running on one host to identify a process running on another host?

2.3 What is Authoritative DNS Server Meaning.